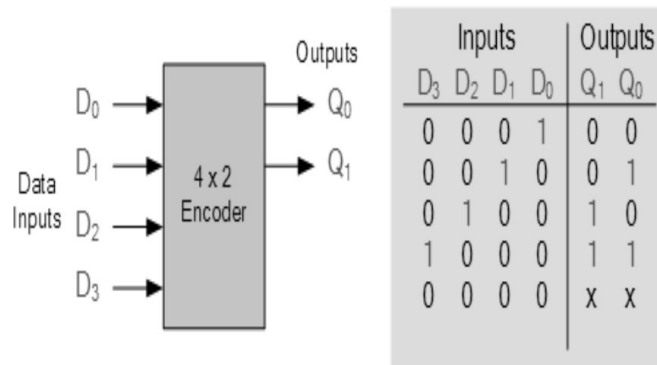


Generic_decoder:

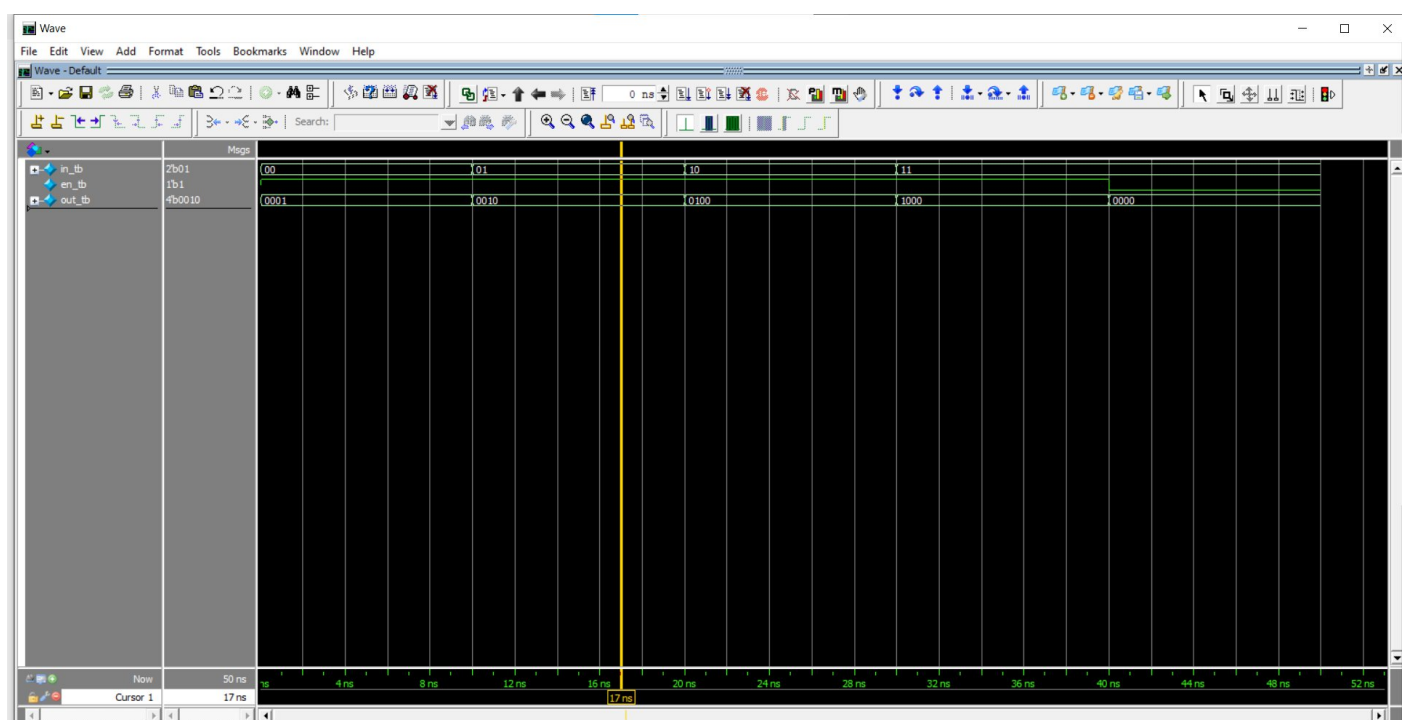
Truth table:



Result:

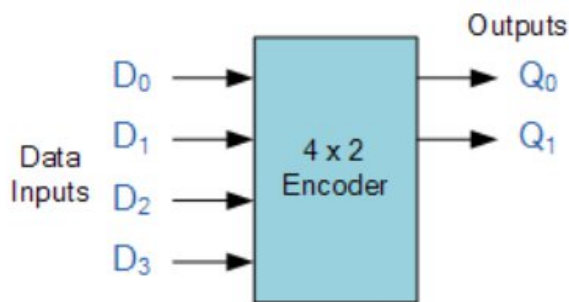
All test cases produced outputs matching the expected values:

```
Library x Project x sim x
Transcript
add wave sim:/decoder_generic_tb/*
VSIM 17> run -all
# t=0 en=1 in=00 out=0001
# t=10 en=1 in=01 out=0010
# t=20 en=1 in=10 out=0100
# t=30 en=1 in=11 out=1000
# t=40 en=0 in=11 out=0000
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/lab 2/decoder_tb.v(23)
#   Time: 50 ns  Iteration: 0  Instance: /decoder_generic_tb
Project : generic  Now: 50 ns  Delta: 0  sim:/de
```



Generic_encoder:

Truth table:



Inputs				Outputs	
D ₃	D ₂	D ₁	D ₀	Q ₁	Q ₀
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1
0	0	0	0	x	x

Result:

All test cases produced outputs matching the expected values:

```
Library x Project x sim x
Transcript
VSIM 20> run -all
# t=0 en=1 in=0001 out=00
# t=10 en=1 in=0010 out=01
# t=20 en=1 in=0100 out=10
# t=30 en=1 in=1000 out=11
# t=40 en=0 in=1111 out=00
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/lab 2/encoder_tb.v(22)
# Time: 50 ns Iteration: 0 Instance: /encoder_generic_tb
Project : generic Now: 50 ns Delta: 0 sim:/er
```

